Digital Forensics Investigation

A DFIR Case Study — Acquisition, File System, Memory, and Network Analysis

Overview

As part of a university digital forensics unit, I completed a multi-stage investigation simulating a full incident-response lifecycle — from evidence acquisition through to network-based attack reconstruction. The exercise used a controlled training dataset; case identifiers, IP addresses, and other potentially sensitive details have been redacted or generalized below. This case study focuses on the methodology and technical skills demonstrated rather than case-specific findings.

Objectives

- Perform and justify a forensically sound acquisition procedure for a storage device.
- Interpret raw disk structures — the Master Boot Record and NTFS $MFT — at the byte level.
- Conduct memory forensics using Volatility to identify process, network, and code-injection artifacts.
- Reconstruct a multi-stage network attack from packet capture data.
- Use a commercial forensic suite (Magnet AXIOM Examine) to correlate artifacts across the registry, browser, and OS layers.

Methodology & Tools

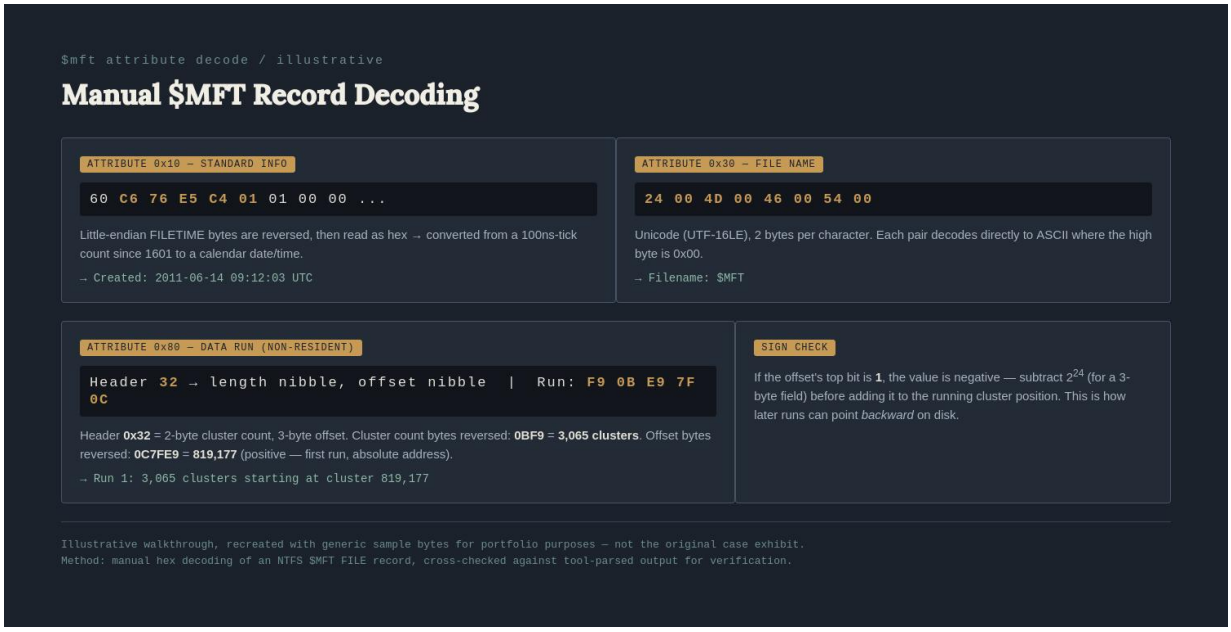
Phase	Tools / Techniques	What I Did
Acquisition	Write-blocking, cryptographic hash verification	Planned a legally admissible acquisition procedure for a removable storage device, weighing physical vs. logical acquisition and justifying physical acquisition with hash verification for evidentiary integrity.
Disk & File System Analysis	Manual hex analysis	Parsed a Master Boot Record by hand to identify partition types, boot flags and sector offsets. Decoded a raw NTFS $MFT record attribute-by-attribute (timestamps, filename, data), manually reconstructing non-resident data runs from hex-encoded cluster offsets.
Memory Forensics	Volatility 3	Enumerated active and terminated processes from a memory image, extracted network connection state, identified processes exhibiting code-injection indicators, and recovered password hashes for credential auditing.

Phase	Tools / Techniques	What I Did
Network Traffic Analysis	Wireshark	Investigated a simulated multi-stage attack: isolated brute-force authentication attempts via HTTP filters, traced an infection chain from an initial script download to a compiled malicious payload, and decoded exfiltrated host data from raw POST traffic.
Artifact Correlation	Magnet AXIOM Examine	Investigated chat, browser and OS-level artifacts. Cross-referenced a Windows registry key with USB device metadata to attribute a removable drive to a specific user account, and used browser session-recovery artifacts to reconstruct undocumented user activity.

Selected Work Samples

The two examples below are recreated with generic sample data for this portfolio — not the original case exhibits — but the technique and reasoning shown are exactly what I applied during the investigation.

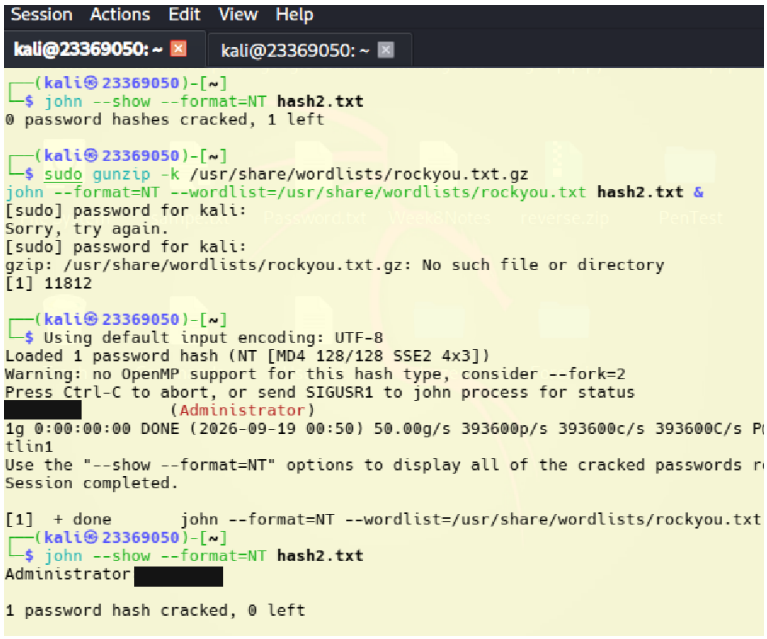




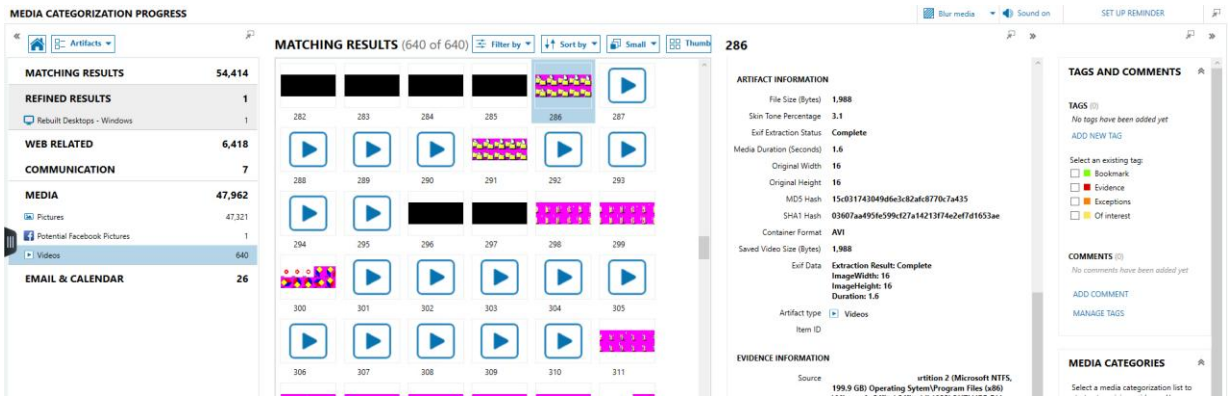
Evidence From My Own Work (Redacted)

The two screenshots below are taken directly from my own assignment work. Case reference numbers, item IDs, and the recovered plaintext value have been blacked out; everything else — commands, tool output, and findings — is exactly as produced.

Credential recovery with John the Ripper



AXIOM filmstrip preview of a recovered video artifact



Key Findings (Generalized)

- Verified that MBR partition metadata correctly identified an active, bootable NTFS partition alongside a Linux-formatted partition and an extended partition.
- Reconstructed the full non-resident data allocation of a system metadata file directly from hex, without relying on tool-parsed output.
- Identified a process exhibiting classic code-injection indicators and correlated it with outbound network activity recorded minutes earlier, supporting a hypothesis of remote compromise.
- Reconstructed an attacker's approach — from credential brute-forcing through malicious script delivery to data exfiltration — using only packet capture evidence.

- Recovered a domain administrator credential hash from a memory image and cracked it using John the Ripper, confirming account compromise beyond mere unauthorized access.
- Identified a process present in a raw process scan but absent from the standard active process listing — a discrepancy consistent with anti-forensic evasion or a since-terminated malicious process.
- Identified acquisition risks specific to solid-state drives, such as TRIM and garbage collection, that can alter data even under write-blocked conditions, and factored this into the acquisition method selected.
- Demonstrated that Windows retains enough metadata (registry and device artifacts) to attribute a specific piece of removable media to a specific user account, even without direct file-access logs.

Skills Demonstrated

- Forensic acquisition planning and chain-of-custody documentation
- Manual binary and hex data-structure analysis (MBR, NTFS $MFT)
- Memory forensics using Volatility 3
- Network traffic analysis and attack reconstruction using Wireshark
- Commercial forensic tool proficiency (Magnet AXIOM Examine)
- Windows registry and cross-artifact correlation
- Credential auditing and password recovery (John the Ripper)

Reflection

This project reinforced that forensic tools accelerate an investigation but don't replace understanding what's happening at the byte level. Manually decoding the $MFT record, for example, made the tool-parsed output in AXIOM meaningfully easier to trust and interpret, because I could verify it against the raw structure myself. It also sharpened how I think about evidentiary integrity — every step, from write-blocking to hash verification, exists to make sure a finding can stand up to scrutiny.

Note: this case study describes coursework completed for a university digital forensics unit using a controlled training dataset. Specific case identifiers, IP addresses, and other potentially sensitive details have been redacted or generalized for confidentiality.